

WASIR KHAN

Mining Engineer | Drilling & Blasting Specialist

مهندس تعدين | أخصائي حفر وتفجير

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Professional Summary

Experienced and safety-driven Drilling and Blasting Engineer with over 7 years of expertise in surface and underground operations across mining, quarrying, hydropower, and tunneling projects. Skilled in designing and implementing efficient, cost-effective, and environmentally responsible blast plans tailored to diverse geological conditions. Proficient in controlled blasting techniques, vibration monitoring, and explosive calculations to ensure structural integrity and minimize environmental impact. Adept in managing drilling and blasting equipment, ensuring strict compliance with HSE regulations and international standards. Known for precision in blast design, operational leadership, and effective coordination with multidisciplinary teams in complex geotechnical settings.

Work Experience

Drilling & Blasting Engineer

Company: FCC Construction Company

Project: NEOM Spine Infrastructure Running Mountains Tunnels, Kingdom of Saudi Arabia
2023 - 2025

JOB RESPONSIBILITIES

- Preparing a face drilling plan using software such as Underground Manager (Epiroc) involves a systematic and detailed process where geological and geotechnical data are carefully analyzed and incorporated into the software to design optimized drilling patterns that consider crucial factors such as rock mechanics, tunnel dimensions, drill hole alignment, and charge placement.
- Developed face drilling patterns and charge layouts based on in-depth geological, geotechnical, and geospatial data analysis.
- Develop detailed blast designs, including drilling patterns, charge placement, and sequencing.
- Monitor blasting activities, analyse results, and submit recommendations to achieve the desired fragmentation.
- Design blast patterns, calculate blast parameters, and determine the amount of explosives required.
- Coordinating with geologists, surveyors, and engineers to understand geological conditions and select appropriate blasting methods.
- Ensuring that all blasting operations comply with safety standards and regulations, while minimizing harmful effects such as vibrations, fly rock, and air-blast; implementing safety procedures for workers and surrounding areas, including blast warnings and evacuation plans.
- Ensure blasting activities minimize environmental impact, including noise, dust, and vibration.
- Maintain and manage blasting equipment and tools, ensuring they are in proper working condition.
- Manage the storage, transportation, and handling of explosive, ensuring compliance with regulations.
- Ensure that all personnel understand and follow safety protocols and operational procedures correctly.

Blasting Engineer

Company: DEV Developer & Infrastructure (Pvt) Ltd

Project: Momand Dam Hydro power Project, Pakistan

2021 - 2022

JOB RESPONSIBILITIES

- Developing safe and efficient drilling and blasting plans, including blast designs, patterns, and sequencing, based on geological data and compliance with safety regulations.
- Designing drilling patterns and determining the precise quantity of explosives required for specific purposes.
- Creating blast patterns that minimize environmental impact, reduce costs, optimize explosive placement, and establish an effective detonation sequence.
- Monitoring safety standards and ensuring strict compliance with local and international regulations regarding the use of explosives, blast design, and operational safety.
- Collecting geological data and analyzing rock characteristics to determine the appropriate quantities of explosives and controlling backbreak during blasting operations.
- Implementing measures to minimize environmental impacts from blasting activities, including controlling ground vibrations, fly rock, and air blast.
- Overseeing blast performance, conducting post-blast analyses to evaluate results, and suggesting adjustments to improve future operations.
- Ensuring proper maintenance and functionality of drilling and blasting equipment, such as drills, explosives-handling tools, and detonators.
- Manage and coordinate the drilling and blasting team, fostering effective communication among departments (e.g., geology, bench planning, and operations) to meet production goals.
- Monitoring and managing the costs associated with drilling and blasting operations, prioritizing efficiency and minimizing waste.
- Maintaining comprehensive records of blast designs, quantities of explosives used, and post-blast outcomes for future reference, performance evaluation, and legal compliance.

Mining Engineer

Company: National Engineering & Transportation Company

Project: Askari Cement Limited Company, Pakistan,

2019- 2021

JOB RESPONSIBILITIES

- Develop effective drilling and blasting plans to ensure optimal fragmentation and material size for the cement manufacturing process.
- Analyze quarry geology and rock characteristics to determine the most suitable explosives and blasting techniques.
- Supervise drilling activities to ensure precise hole positioning, depth, and alignment for efficient blasting.
- Execute safe and controlled blasting operations, adhering to all safety regulations and environmental standards.
- Ensure compliance with local, national, and international regulations related to drilling and blasting in quarry operations.
- Monitor and manage resources to achieve quarry production targets and maintain raw material supply for the cement plant.
- Minimize environmental impacts of blasting, such as vibration, noise, and dust emissions, by employing best practices.
- Coordinate with other departments, such as production and maintenance teams, to ensure uninterrupted material flow to the cement plant.
- Evaluate blasting results and implement improvements to enhance fragmentation, reduce costs, and increase overall efficiency.

Mining Engineer

Company: Ghulam Rasool & Company (Pvt) Ltd

Project: Jabori Hydro Power Project, Pakistan

2017- 2019

JOB RESPONSIBILITIES

- Develop and implement effective drilling and blasting strategies to achieve optimal rock fragmentation and material size suitable for cement manufacturing processes.
- Conduct detailed geological and geotechnical assessments to select appropriate explosives and tailor blasting techniques to varying rock formations.
- Supervise drilling operations to ensure precise borehole alignment, spacing, and depth in accordance with blast design specifications.
- Execute controlled and safe blasting operations while ensuring full compliance with local, national, and international safety and environmental regulations.
- Monitor blast performance and analyze post-blast results to identify improvement areas for fragmentation quality and cost-efficiency.
- Manage explosive inventory, storage, and handling in alignment with legal safety standards and industry best practices.
- Minimize environmental impacts such as ground vibration, air-blast, noise, and dust through the application of advanced mitigation techniques.
- Collaborate with production, maintenance, and quality control teams to ensure consistent raw material supply to the cement plant.
- Optimize resource allocation, including labor, machinery, and explosive materials, to meet daily and monthly production targets.
- Drive continuous improvement initiatives by integrating new technologies and refining blast parameters to enhance operational performance and reduce costs.

Education

Balochistan University of Information Technology, Engineering & Management Sciences

Bachelor of Science in Mining Engineering

2013 - 2017

Certifications

- Institution of Occupational Safety and Health (IOSH) - Safety Management
- Occupational Safety and Health Administration (OSHA) - Workplace Safety
- Confined Space Entry & Rescue Procedures
- Risk Assessment & Work at Height Protocols
- FCC Construction Company - Certificate of Commendation

Technical Proficiency

Blasting & Geotechnical Software:

ShotPlus | ViewShot | Underground Manager (Epiroc) | O-Pit Blast

Design & Documentation:

AutoCAD | MS Office (Word, Excel, PowerPoint)

Core Competencies

- **Mine Planning & Design** – Expertise in short-term and long-term mine planning using modern software.
- **Drilling & Blasting Operations** – Proficient in designing safe and effective blast patterns and explosive usage.
- **Rock Mechanics & Ground Control** – Strong understanding of geotechnical principles to ensure stability and safety.
- **Mineral Exploration & Resource Estimation** – Skilled in geological data interpretation and reserve modeling.
- **Mine Ventilation & Water Management** – Experience in designing efficient ventilation and drainage systems.
- **Health, Safety & Environmental Compliance (HSE)** – Well-versed in OSHA, ISO, and local mining regulations.
- **Production Monitoring & Optimization** – Ability to increase operational efficiency and reduce waste.
- **Cost Control & Budget Management** – Competent in analyzing and managing mining project budgets.
- **Explosive Management & Regulatory Compliance** – Knowledge of explosive handling, storage, and safety protocols.
- **Project Management** – Experience leading mining operations, from site development to closure.
- **Technical Report Writing** – Skilled in preparing feasibility studies, compliance documents, and operational reports.
- **Team Leadership & Communication** – Proven ability to manage diverse teams in multicultural environments.
- **Risk Assessment & Mitigation** – Identifying hazards and implementing proactive safety measures.
- **Sustainability & Environmental Stewardship** – Commitment to minimizing environmental impact in operations.
- **Blasting Vibration & Fly Rock Control** – Minimizing off-site impacts from blasting operations.
- **Production Planning & Optimization** – Maximizing efficiency and output through detailed operational planning.
- **Risk Assessment & Hazard Control** – Identifying potential risks and implementing control measures.
- **Material Handling & Transport Systems** – Designing efficient ore and waste transport processes.
- **Process Improvement & Lean Mining** – Streamlining operations to reduce waste and increase productivity.
- **Team Leadership & Workforce Coordination** – Supervising and aligning team efforts to meet operational goals.
- **Conflict Resolution & Team Motivation** – Maintaining harmony and motivation in multicultural work environments.

Pakistan Engineering Council

Pakistan Engineering Council (PEC) – Registered Mining Engineer (**MINING/01988**)

Driving License

Valid Private Vehicle License – Kingdom of Saudi Arabia

Languages Spoken

- English – Fluent | Professional spoken and written communication
- Urdu (اُردُو) – Native/Fluent | Excellent in formal and creative expression
- Hindi (हिन्दी) – Fluent | Skilled in both written and verbal communication
- Arabic (العربية) – Proficient | Strong working knowledge in speaking communication.

Reference

References will be provided upon request.

